KAIZEN MATHS

Lesson starter

Function Notation, Composite and Inverse Functions

Numerical Substitution

Name _____

S1

For f(x) = -x - 6, find f(2).

S2

For $f(x) = x^2 - 3x - 5$, find f(3).

S3

For f(x) = -6x, find f(5).

S4

For $f(x) = 2x^2 + 3$, find f(3).

Show your working

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Show your working

KAIZEN MATHS

Exit ticket

Function Notation, Composite and Inverse Functions

Numerical Substitution

Name _____

E1

For f(x) = 2x + 3, find f(1).

E2

For $f(x) = x^2 - x + 2$, find f(0).

E3

For f(x) = -x - 9, find f(-1).

KAIZEN MATHS

Exit ticket

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Numerical Substitution

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E1

For f(x) = 2x + 3, find f(1).

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For $f(x) = x^2 - x + 2$, find f(0).

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For f(x) = -x - 9, find f(-1).

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Show your working

KAIZEN MATHS

Exit ticket

Function Notation, Composite and Inverse Functions

Numerical Substitution

Name _____

E1

For f(x) = 2x + 3, find f(1).

E2

For $f(x) = x^2 - x + 2$, find f(0).

E3

For f(x) = -x - 9, find f(-1).

ALG-2F88-L1 · E

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Function Notation, Composite and Inverse Functions

Numerical Substitution

Name _____

E1

For f(x) = 2x + 3, find f(1).

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For $f(x) = x^2 - x + 2$, find f(0).

E3

For f(x) = -x - 9, find f(-1).

ALG-2F88-L1 · E

Show your working

Teacher answers & working

Function Notation, Composite and Inverse Functions · Numerical Substitution

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S1

For f(x) = -x - 6, find f(2).

Answer:

-8

Working:

Step 1: The input is 2, so replace every x in f(x) with 2.

$f(2) = -1(2) - 6$

Step 2: Work out the multiplication first, then combine the constant term.

$f(2) = -2 - 6$

$f(2) = -8$

S2

For f(x) = x^2 - 3x - 5, find f(3).

Answer:

-5

Working:

Step 1: Replace every x with 3 and use brackets for the substituted value.

$f(3) = 1(3)^2 - 3(3) - 5$

Step 2: Square the input before multiplying by the coefficient.

$f(3) = 9 - 9 - 5$

Step 3: Combine the numerical terms.

$f(3) = -5$

S3

For f(x) = -6x, find f(5).

Answer:

-30

Working:

Step 1: The input is 5, so replace every x in f(x) with 5.

$f(5) = -6(5)$

Step 2: Work out the multiplication first, then combine the constant term.

$f(5) = -30$

$f(5) = -30$

Teacher answers & working

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S4

For f(x) = $2x^2 + 3$, find f(3).

Answer:

21

Working:

Step 1: Replace every x with 3 and use brackets for the substituted value.

$f(3) = 2(3)^2 + 0(3) + 3$

Step 2: Square the input before multiplying by the coefficient.

$f(3) = 18 + 3$

Step 3: Combine the numerical terms.

$f(3) = 21$

E1

For f(x) = 2x + 3, find f(1).

Answer:

5

Working:

Step 1: The input is 1, so replace every x in f(x) with 1.

$f(1) = 2(1) + 3$

Step 2: Work out the multiplication first, then combine the constant term.

$f(1) = 2 + 3$

$f(1) = 5$

E2

For f(x) = x^2 - x + 2, find f(0).

Answer:

2

Working:

Step 1: Replace every x with 0 and use brackets for the substituted value.

$f(0) = 1(0)^2 - 1(0) + 2$

Step 2: Square the input before multiplying by the coefficient.

$f(0) = 0 + 2$

Step 3: Combine the numerical terms.

$f(0) = 2$

Teacher answers & working

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E3

For f(x) = -x - 9, find f(-1).

Answer:

-8

Working:

Step 1: The input is -1, so replace every x in f(x) with -1.

$f(-1) = -1(-1) - 9$

Step 2: Work out the multiplication first, then combine the constant term.

$f(-1) = 1 - 9$

$f(-1) = -8$